

Requirements

1. Customers can browse books, view details, and place orders.
2. Each book can have one or multiple authors.
3. Customer place orders, which can include multiple books.
4. Each order has a payment associated with it.
5. Customers can leave reviews for each book they purchased.
6. Each order can be assigned to only one staff member who processes the purchase.

Entities

Entity 1: Customer

The requirements state that customers can browse books, place orders, and leave reviews. Since customers perform several important actions, a separate Customer entity is required.

Attributes

- CustomerID (Primary Key)
- FirstName
- LastName
- Email
- Phone
- Address

Entity 2: Book

Books are the primary products sold by the bookstore. Every order, review, and author relationship revolves around books.

Attributes

- BookID (Primary Key)
- Title
- Genre
- Price
- PublicationYear
- StockQuantity

Entity 3: Author

The requirements specify that one book may have multiple authors. Instead of storing author names directly inside the Book table, creating a separate Author entity eliminates duplicate author information.

Attributes

- AuthorID (Primary Key)
- AuthorName
- Biography

Entity 4: Order

Orders represent purchases made by customers. Every purchase needs to store information such as purchase date, total amount, and who placed the order.

Attributes

- OrderID (Primary Key)
- OrderDate
- TotalAmount
- CustomerID (Foreign Key)
- StaffID (Foreign Key)

Entity 5: OrderItem

An order can contain multiple books, and a book can appear in multiple orders.

This creates a **many-to-many relationship** between Order and Book, resolved using OrderItem.

Attributes

- OrderItemID (Primary Key)
- OrderID (Foreign Key)
- BookID (Foreign Key)
- Quantity
- UnitPrice

Entity 6: Payment

Instead of storing payment information inside the Order table, creating a separate Payment entity makes the database easier to expand if additional payment details are needed later.

Attributes

- PaymentID (Primary Key)
- PaymentDate
- Amount
- PaymentMethod
- PaymentStatus
- OrderID (Foreign Key)

Entity 7: Review

A Review entity stores customer feedback separately from books because many customers can review the same book.

Attributes

- ReviewID (Primary Key)
- Rating
- Comment
- ReviewDate
- CustomerID (Foreign Key)
- BookID (Foreign Key)

Entity 8: Staff

Staff members process orders.

Attributes

- StaffID (Primary Key)
- StaffName
- Email
- Position

Entity 9: BookAuthor

Since a book may have multiple authors and an author may write multiple books, a many-to-many relationship exists.

Attributes

- BookID (Foreign Key)
- AuthorID (Foreign Key)

Composite Primary Key

- (BookID, AuthorID)

Relationships

Customer → Order

- One customer can place many orders.
- One order belongs to one customer.

Relationship: 1:M

Order → Payment

- Each order has exactly one payment.

- Each payment belongs to one order.

Relationship: 1:1

Order → OrderItem

- One order contains many order items.
- Each order item belongs to one order.

Relationship: 1:M

Book → OrderItem

- One book can appear in many order items.
- Each order item references one book.

Relationship: 1:M

Book ↔ Author

- A book can have multiple authors.
- An author can write multiple books.

Relationship: M:N

Resolved through **BookAuthor**.

Customer → Review

- One customer can write many reviews.
- Each review is written by one customer.

Relationship: 1:M

Book → Review

- One book can have many reviews.
- Each review belongs to one book.

Relationship: 1:M

Staff → Order

- One staff member can process many orders.
- Each order is processed by only one staff member.

Relationship: 1:M

